IT457 CLOUD COMPUTING • CASE STUDY PLAN

CoWIN: National-Scale Vaccination Scheduling Cloud Platform

Architectural Extraction, Infrastructure Economics, and Outcome Analysis (Week 1 Checkpoint)

Track: Govt & Public Digital Platforms
Scale: 1.38B Citizens / National Scale
Date: Semester Coursework Checkpoint

2.20B+	25.1M	3.2B+	< 0.3%
DOSES CERTIFIED	SINGLE-DAY PEAK RECORD	DAILY PEAK API INGRESS	PLATFORM COST OF TOTAL OUTLAY

1. CASE STUDY SALIENT POINTS

A. PROBLEM CONTEXT & CORE SCALING BOTTLENECKS

- **Thundering-Herd Slot Stampede:** Millions of concurrent citizens queried and reserved slots the instant regional quotas opened, inducing severe DB lock contention and thread starvation.
- **Write-Heavy Mutation Workload:** Required thousands of guaranteed, non-duplicating transactional writes/sec (inventory decrement, vaccinator confirmation) under strict ACID integrity.
- **Inclusivity & Offline Operations:** Supported walk-in assisted registrations across rural centers over 2G/3G networks, issuing tamper-proof credentials without continuous connectivity.
- **Sovereign Data Governance:** Citizen health and ID records strictly localized within sovereign cloud borders, prohibiting data transit outside Indian jurisdiction.

B. COST DATA & INFRASTRUCTURE ECONOMICS

- **Ultra-Frugal Software & Cloud Spend:** Platform engineering and cloud infrastructure cost an estimated **₹50–₹100 Cr (~$7–$13M)**, representing <0.3% of the ₹35,000 Cr vaccine procurement budget.
- **Global Benchmarking Advantage:** Far lower TCO than international systems (UK NHS App >£300M, US CDC VAMS ~$44M). Amortized cost was under **₹0.50 ($0.006) per beneficiary**.
- **Hybrid Cloud Architecture:** Core databases and citizen vaults sit on National Informatics Centre (NIC/MeghRaj) cloud with zero-egress public network tariffs; AWS India provided elastic burst nodes for peak mega-drives.
- **Decentralized Cryptographic Savings:** Digital QR codes with Ed25519 signatures offloaded verification checks to edge verifiers, saving petabytes of central server bandwidth.

C. EXTRACTED CLOUD ARCHITECTURE BLUEPRINT

Edge & Ingress (Anti-Stampede)
Route 53 Anycast DNS + CloudFront CDN edge PoPs. AWS WAF and API Setu enforce rate-limiting (100 req/5 min/IP) to deflect scrapers and bots from backend origin nodes.

Orchestration & Microservices
Decoupled stateless microservices on Kubernetes (EKS & NIC MeghRaj). Auto-scales dynamically via custom metrics (queue backlog depth and CPU/memory pressure).

Distributed Caching (Concurrency)
Partitioned Redis (ElastiCache) clusters hold district/pincode slot quotas. Atomic Redis operations (DECR / Redlock) absorb high-frequency slot booking bursts prior to DB writes.

Polyglot Persistence Layer
PostgreSQL manages ACID inventory/booking commits; MongoDB/DocumentDB handles flexible citizen profile lookups; ClickHouse powers sub-second real-time admin analytics.

Asynchronous Event Pipelines
Apache Kafka & Amazon SQS event streams decouple write flows: asynchronous OTP dispatch across telecom gateways, vaccination logging, and background PDF certificate generation.

Verification & Open DPG (DIVOC)
Built upon DIVOC & Sunbird RC open standards. W3C-compliant Verifiable Credentials with asymmetric Ed25519 key signing enable instant offline QR validation without central DB hits.

Week 3 Exercise Plan Alignment (Govt Platforms Track): Per the syllabus, we will design a multi-region, high-availability architecture and build a **High-Concurrency Distributed Slot Reservation Engine** replicating CoWIN's concurrency pipeline: containerized microservice behind an ALB, Redis distributed lock manager (atomic inventory decrements), Kafka/RabbitMQ async event buffer, and PostgreSQL transactional persistence. We will evaluate latency and scaling behaviors under simulated thundering-herd loads using Locust/k6.

2. PRIMARY SOURCES & CASE STUDY CITATIONS FOUND

CATEGORY	PRIMARY SOURCE / DOCUMENT	ISSUING BODY / AUTHOR	KEY TECHNICAL / QUANTITATIVE INSIGHT EXTRACTED
GOVERNMENT & SPECS	**CoWIN Portal & Open API Specifications** *apisetu.gov.in / co-vin.in*	Ministry of Health (MoHFW) & National Health Authority (NHA)	Rate-limiting rules, tokenized beneficiary APIs, multi-tenant center queries, walk-in vaccinator sync schemas.
LEADERSHIP / ARCH	**"CoWIN: Technology Behind World's Largest Vaccination Drive"**	Dr. R.S. Sharma (CEO, NHA / Head, Empowered Group)	Architectural principles: Single source of truth, zero geo-tracking privacy by design, stateless microservices scaling.
CORE ENGINE CODE	**DIVOC Open-Source Architecture & Stack** *github.com/egovernments/divoc*	eGovernments Foundation & DPGA	Production deployment configs with Keycloak (IdP), Apache Kafka streaming, ClickHouse OLAP, and W3C JSON-LD credentialing.
CLOUD CASE STUDY	**"Scaling CoWIN Platform on AWS Cloud"** & Intel Xeon Optimization Brief	Amazon Web Services (AWS) & Intel Corporation	Kubernetes (EKS) auto-scaling, ElastiCache Redis write optimization, CloudFront CDN edge caching, and cryptographic compute acceleration.
ECONOMICS / POLICY	**"Stacking Up the Benefits: Lessons from India's Digital Journey"** (WP/23/78)	International Monetary Fund (IMF) & World Bank	Quantified DPI economics, operational frugality (<0.3% of total vaccination program budget), and Aadhaar/DigiLocker integration.